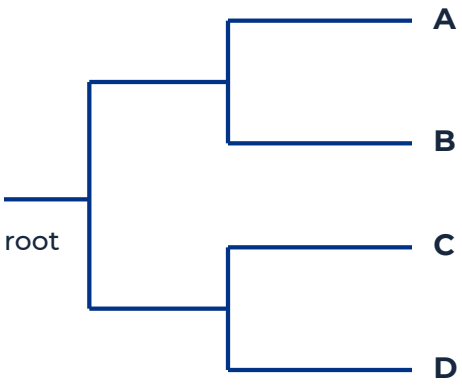
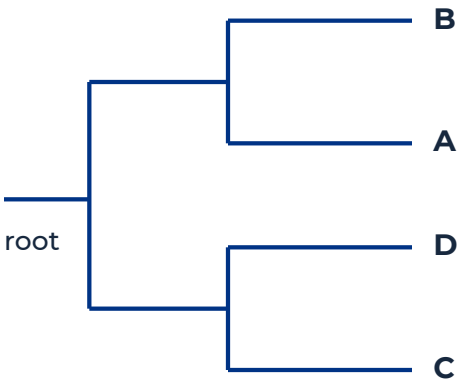
Phylogenetic tree guide

Read shared ancestry from branching order, not tip spacing or page position.

Same relationships





Read the tree	What it tells you
Sister taxa	A-B and C-D share their own most recent common ancestors in both drawings.
Clade	An ancestor and all its descendants; C and D form one clade in this example.
Node rotation	Rotating a branch point changes drawing order, not the inferred relationships.
Common ancestry	A is closer to B than C by recency of common ancestry, not by ruler distance between labels.
Branch length	Interpret time or amount of change only when the diagram states what length means.

Common pitfalls: "higher" tips are not more evolved; living tips are not automatically ancestors of other tips. These hypothetical rooted trees show no time scale, fossils or uncertainty.

Phylogenetic tree guide

Self-check questions and explanations. Refer to the visual guide on the front.

Are A and B or A and C more closely related in this rooted tree?

A and B share the upper branching node; A and C share the root-side node joining both major clades. A-B is the more recent pair in this rooted topology.

Do the two rotated trees show the same relationships?

Yes. Sister pairs are A-B and C-D in both. Rotating branches around an internal node changes tip order but preserves ancestry and clade membership.

Is a tip higher on the page more evolved?

No. Each terminal tip represents a sampled lineage/taxon; vertical placement is a drawing choice. Living tips are not ranked by how evolved they are.

If E is sister to C, what is the revised topology?

Revised relationship: ((A,B),((C,E),D)). C and E form the smallest two-tip clade; C,D,E form the next clade. The ancestor joining C-E branches is nested within the C-D-E ancestor.

If a trait originated once on the C-D clade stem with no losses, does E inherit it?

C, D and E inherit the trait. Place its origin on the stem of their shared clade before it splits, given the once/no-loss assumption. Physical spacing is unrelated to inheritance.

Can arbitrary branch length show elapsed time?

Need a stated branch-length meaning and an appropriate time scale/calibration or dated model. In this diagram lengths are arbitrary; other trees may represent time or amount of inferred change.